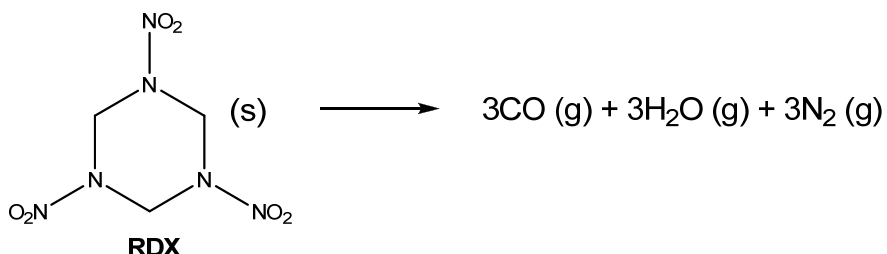


2018 H1 Chem Prelim P2 (Suggested Solutions)

- 1 The explosive RDX is a white crystalline solid first prepared in 1899 but wasn't used as an explosive until 1920 by a German chemist called Herz. He prepared RDX by direct nitration of hexamine and found it be very sensitive to denotation by impact or friction. During World War 2, it was added to TNT to increase its explosive power. The explosive decomposition of RDX is as shown by the equation below.



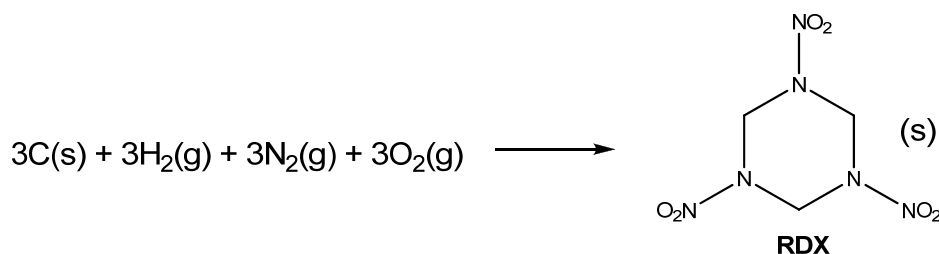
- (a) (i) Calculate the volume of gases, in cm³, produced at s.t.p during the explosive decomposition of 2.22 g of RDX given that the relative molecular mass of RDX [2] is 222.

$$\text{Moles of RDX} = 2.22 / 222 = 0.01 \text{ mol}$$

$$\text{Moles of gas formed} = 9 \times 0.01 = 0.09 \text{ mol}$$

$$\text{Volume of gases} = 0.09 \times 22.7 = 2.04 \text{ dm}^3 = 2040 \text{ cm}^3$$

- (ii) Write an equation to represent the standard enthalpy change of formation of [1] RDX.



State symbol needed

- (iii) Use the enthalpies of formation shown below to calculate the standard enthalpy change for the explosive decomposition of RDX.

Compound	$\Delta H_f / \text{kJ mol}^{-1}$
RDX (s)	+ 62
CO (g)	- 111
H ₂ O (g)	- 242

[2]

$$\Delta H_{\text{decomposition}} = [3(-111) + 3(-242)] - 62]$$
$$= - 1121 \text{ kJ mol}^{-1}$$

- (iv) Scientific literature usually quote the enthalpy change of formation for chemical explosives in kJ kg⁻¹. Calculate the enthalpy change of formation of RDX from kJ mol⁻¹ to kJ kg⁻¹. [1]

$$1 \text{ mole} = +62 \text{ kJ}$$

Hence, 222 g = +62 kJ

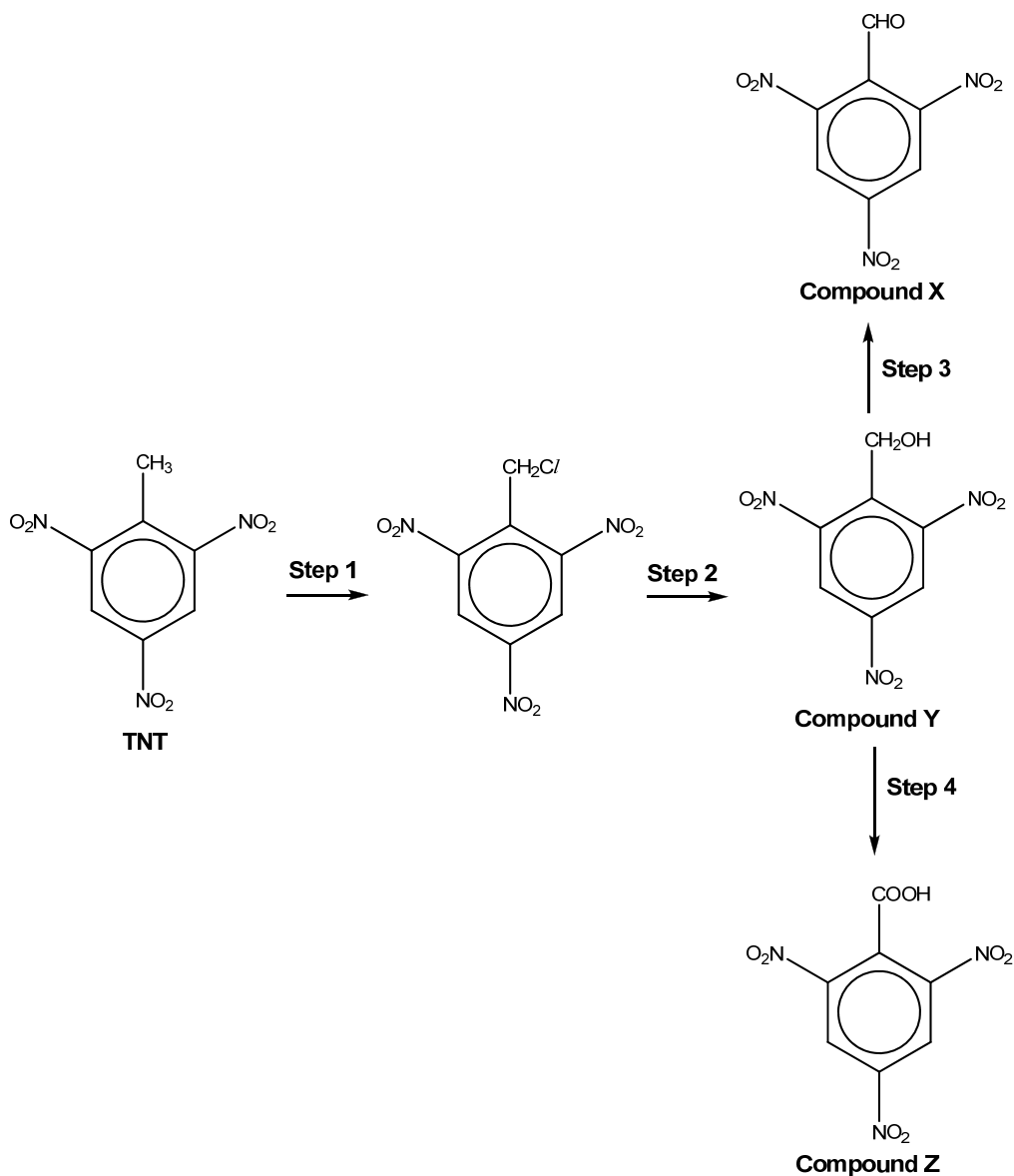
1 kg = + 279 kJ

ΔH_f of RDX = + 279 kJ kg⁻¹

- (v) With reference to your answer in (iii), suggest if RDX is more stable than the products of decomposition. [1]

RDX is less stable than the products.

- (b) The reaction scheme below shows the reactions involving TNT.



- (i) State the reagent and condition for Steps 1 – 4. [4]

Step 1: limited chlorine, UV light

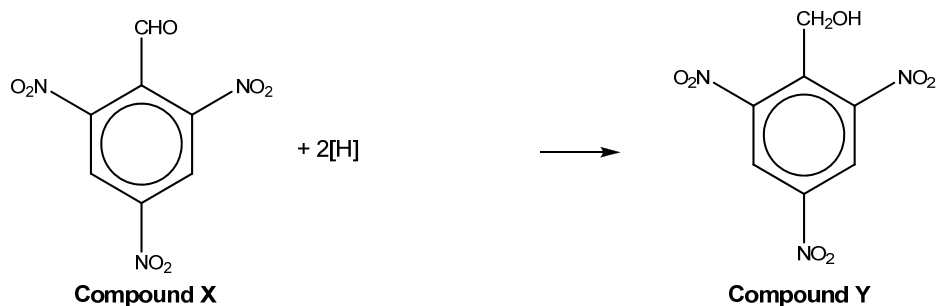
Step 2: NaOH (aq) heat

Step 3: K₂Cr₂O₇, H₂SO₄ (aq)/dilute/ heat with distillation

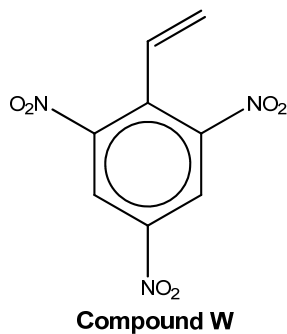
Step 4: KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$, H_2SO_4 (aq)/dilute heat

- (ii) Besides lithium aluminium hydride, state the reagent and condition that can reduce compound **X** into compound **Y**. Write a balanced equation for this reaction. [2]

NaBH_4 or hydrogen gas with nickel/ platinum, heat



- (iii) Compound **Z** can also be synthesised by the oxidation of the following compound **W**.



State and explain if the above compound exhibits cis-trans isomerism. [1]

Compound **W** does not exhibit cis-trans isomerism as the carbon of the double bond is connected to the same atom.

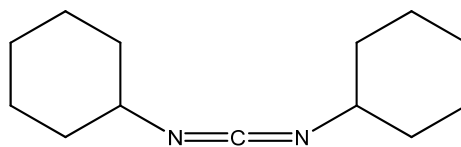
- (iv) Suggest a suitable chemical test to distinguish between Compound **W** and TNT. [2]

Reagent and condition: Bromine liquid (dark)

Observation: orange-red will turn colourless for Compound **W**. orange-red colour will remain for TNT.

[Total: 16]

- 2 N,N -dicyclohexylcarbodiimide often abbreviated as DCC, is used to couple amino acids during artificial peptide synthesis. The structure of DCC is as shown below.



DCC

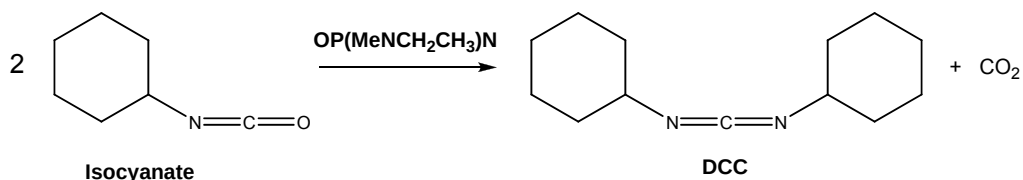
- (a) (i) Define the term *relative molecular mass*. [1]

The relative molecular mass (M_r) of a substance is defined as the ratio of average mass of one molecule of the substance to $\frac{1}{12}$ the mass of an atom of ^{12}C isotope based on the ^{12}C scale.

- (ii) Calculate the relative molecular mass of DCC to 1 decimal place. [1]

$\text{C}_{13}\text{H}_{22}\text{N}_2$. Relative molecular mass = 206.0

- (b) DCC can also be prepared by reacting isocyanate in the presence of a catalyst, $\text{OP}(\text{MeNCH}_2\text{CH}_3)_3$, as shown by the following equation. The yield of the reaction is typically 92.0%.



- (i) A student prepared DCC via the above synthesis method by using 0.100 g of isocyanate and 0.035 g of DCC was produced. Calculate the percentage yield of his experiment and hence suggest if the student's experimental technique is accurate given that the M_r of isocyanate is 125.0. [2]

Mole of isocyanate = $0.100 / 125 = 8 \times 10^{-4}$ mol

Mole of DCC = 4×10^{-4} mol

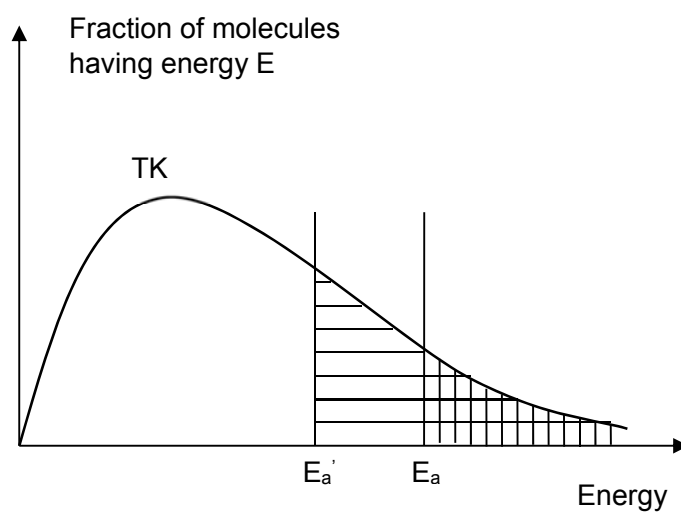
Theoretical yield of DCC = $4 \times 10^{-4} \times 206 = 0.0824$ g

% yield of DCC = $0.035 / 0.0824 \times 100\% = 42.5\%$

(Since % yield is < 92.0%, hence the student's experimental technique is inaccurate.

- (ii) Explain, with the aid of an energy distribution diagram, how the addition of $\text{OP}(\text{MeNCH}_2\text{CH}_3)_3$ affects the rate of the above chemical reaction. [3]

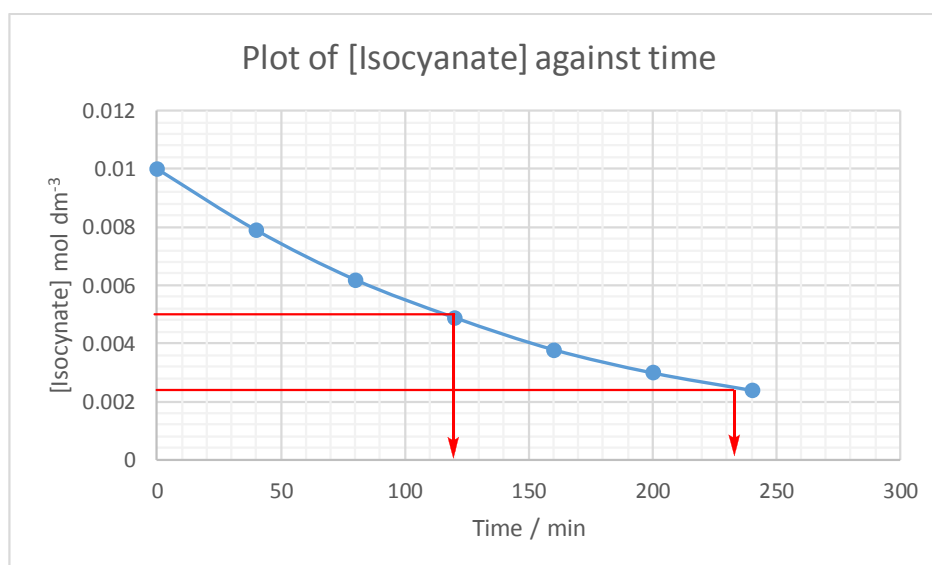
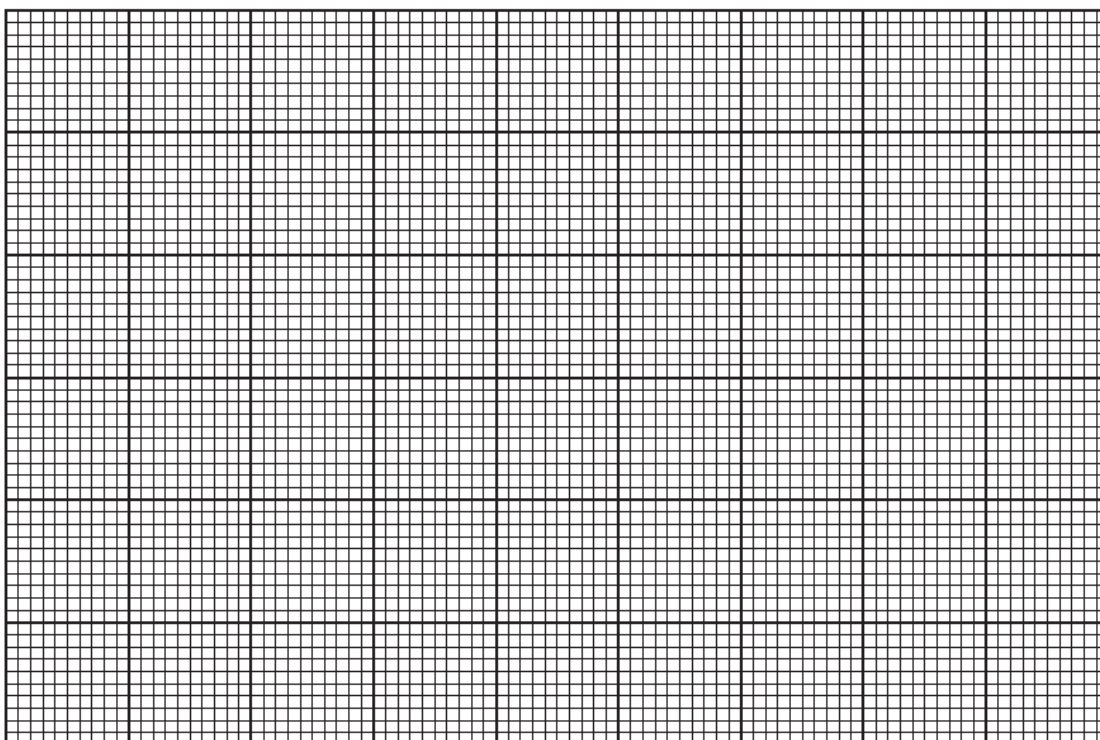
- A catalyst lowers the activation energy of a reaction by providing an alternative reaction pathway.
- Increase in the number of molecules having energy greater than or equal to the lowered activation energy.
- Frequency of effective collisions increases.
- Reaction rate thus increases.
- Diagram



- (iii) The progress of the reaction can be followed by plotting the concentration of isocyanate against time. Plot the experimental results on the grid below to determine the rate equation for this reaction.

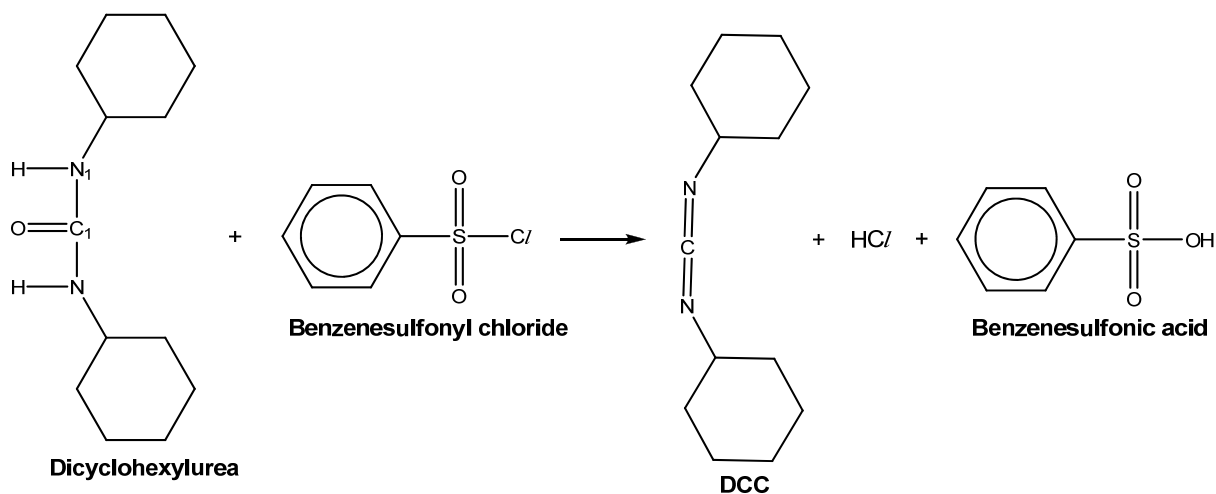
Time/ min	[Isocyanate] / mol dm ⁻³
0	0.0100
40	0.0079
80	0.0062
120	0.0049
160	0.0038
200	0.0030
240	0.0024

[3]



- Plot of graph with axes (and units)
- 1st $t_{1/2}$ = 116 min; 2nd $t_{1/2}$ = 114 min $\Rightarrow$ Average $t_{1/2}$ = 115 min show on graph or working
- Since $t_{1/2}$ is a constant, order of reaction is 1st order wrt [isocyanate].
Rate = k [isocyanate]

(c) DCC can also be synthesised from dicyclohexylurea and benzenesulfonyl chloride as shown in the following equation.

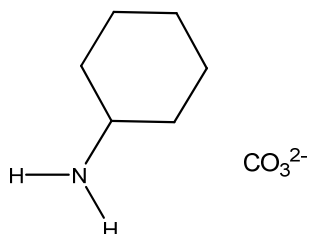


- (i) State the VSEPR (valence shell electron pair repulsion) theory to predict the shape about N₁ and C₁ in dicyclohexylurea. [3]

Electron pairs **repel** each other and arrange themselves as far apart as possible to maximise stability and minimise electrostatic repulsion.

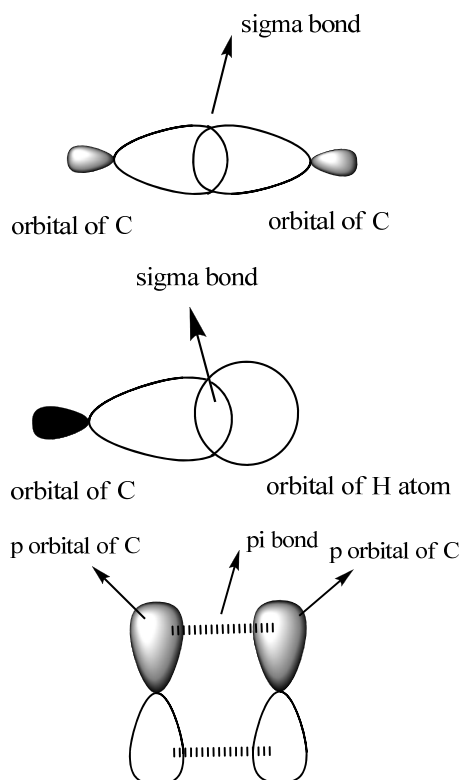
The shape is trigonal pyramidal about N₁ as there are 3 bond pairs and 1 lone pair. The shape is trigonal planar about C₁ as there are 3 bond pairs.

- (ii) Draw the structural formulae of the products formed when dicyclohexylurea reacts with hot aqueous sodium hydroxide. [2]



- (iii) Benzenesulfonic acid has a benzene ring. Describe the bonding in benzene in terms of orbital overlap. Draw relevant diagrams to illustrate your answer. [4]

- Each carbon atom will overlap head on with 2 neighbouring carbon atom as well as the s orbital of one H atom to form 3 σ (sigma) bonds
- Each carbon will also have an p orbital that would overlap side-on, both above and below the ring of carbon atoms, with the p orbitals of the 2 neighbouring carbon atom, forming a delocalised ring of π (pi) electrons

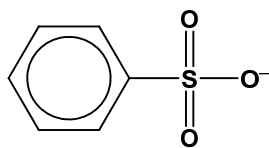


- (iv) Weak acids typically have a K_a value that is less than 1. The pK_a value of benzenesulfonic acid is -2.8. Deduce if benzenesulfonic acid is a weak acid. [1]

The K_a of benzenesulfonic acid is $10^{2.8}$, which is **more than one**. Hence, benzenesulfonic acid is not a weak acid.

- (v) Define the term conjugate base. Hence, draw the structural formula for the conjugate base of benzenesulfonic acid. [2]

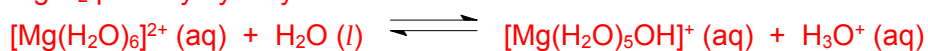
Conjugate base refers to a base differing from its acid by 1 H⁺.



- (vi) Benzenesulfonic acid can be converted into benzenesulfonyl chloride by using a period 3 chloride such as phosphorus pentachloride. Two other chlorides of period 3 elements include silicon chloride and magnesium chloride. Using the chlorides of magnesium, silicon and phosphorus as examples, describe the reactions of the chlorides of the third period of the Periodic table with water, writing equations where appropriate. [5]

Hence, suggest what influence the type of bonding present in these chlorides has on their reaction with water.

MgCl₂ partially hydrolyses in water



SiCl₄ and PCl₅ undergo complete hydrolysis in water.



OR

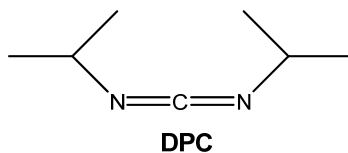


MgCl₂ has ionic bonds with covalent character

Chlorides of Si and P have covalent bonds.

As the covalent character of the chlorides increase, the more complete the hydrolysis in water hence producing a more acidic solution.

- (d) N,N-diisopropylcarbodiimide, DPC, was developed as an alternative to DCC. DPC is identical to DCC in nearly every way except that DPC is less likely to cause an allergic reaction. Furthermore, as a liquid, DPC is easier to handle than DCC which is a waxy solid. Explain the difference in physical states between DCC and DPC.

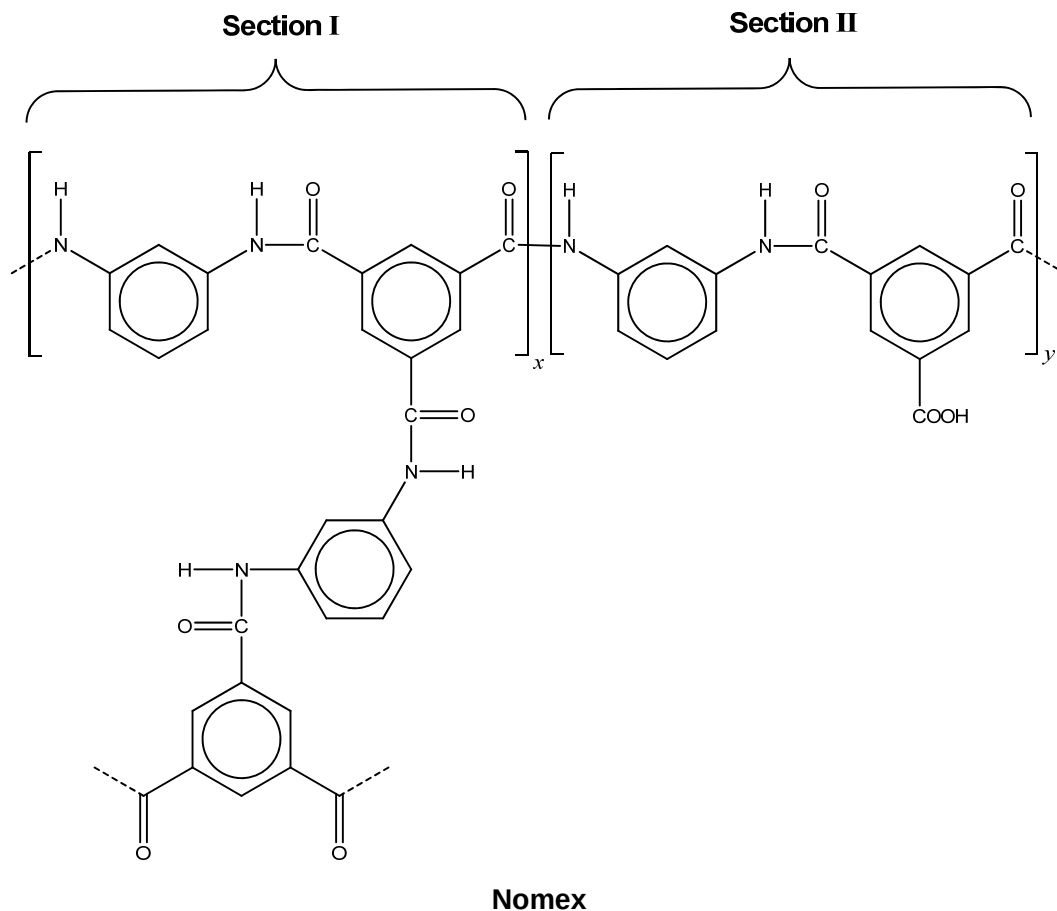


[2]

Both DPC and DCC are polar simple covalent molecules. DPC has a smaller electron cloud than DCC hence the instantaneous dipole-induced dipole interactions between DPC is weaker. Hence, less energy is required to overcome the intermolecular forces of attraction between DPC than DCC. Hence, DCC will exist as a solid due to the stronger intermolecular forces of attraction.

[Total: 29]

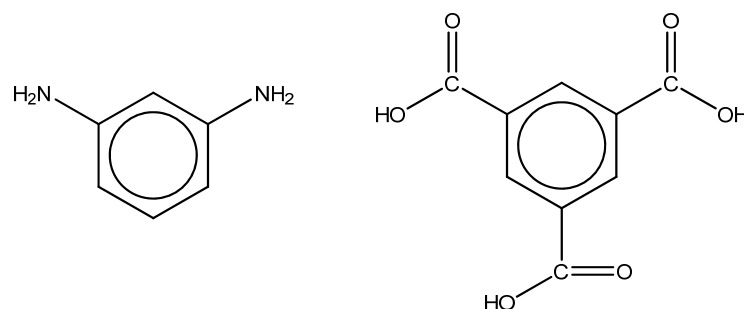
- 3 Nomex is a class of aramid fibres used as a heat-resistant and strong synthetic fibres in aerospace and military applications. Two sections of the polymer, Nomex, is as shown below.



- (a) Define polymer. [1]

Polymers are macromolecules built up from monomers, with average molar mass of at least 1000 or at least 100 repeat units.

- (b) (i) With reference to Section II, draw the structural formula of the monomers in Nomex. Hence, state the type of polymerisation involved. [3]

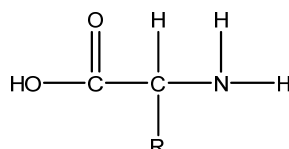


- (ii) Classify the polymer represented by Section I. Hence, use its structure to explain a property of this polymer. [3]

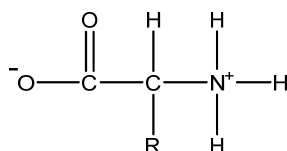
Section I : Thermoset/cross-linked

Thermoset: It does not melt / soften / decomposes / lose its shape on heating due to strong covalent bonds holding the chains together which does not break upon heating.

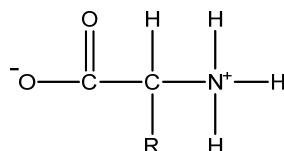
- (c) Wool is a texture fibre obtained from sheep and other animals. Wool consists of a polymer chain constructed from amino acids. The general structure of an amino acid is as shown below.



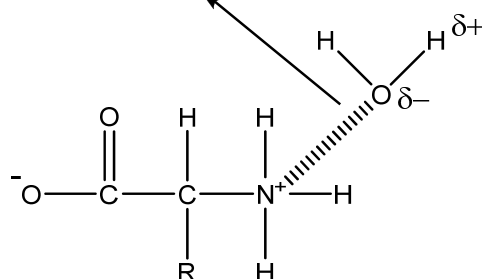
In water, amino acids forms zwitterions as shown below.



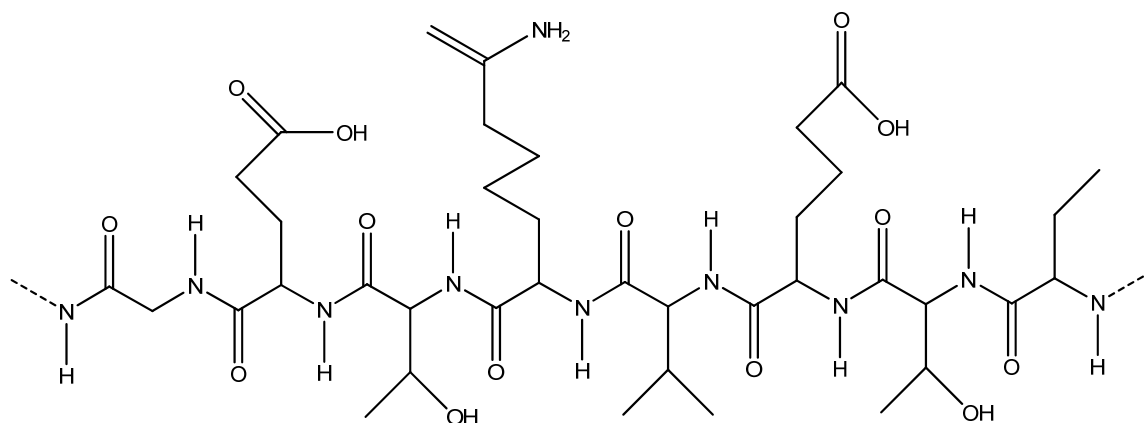
- (i) Hygroscopy is the phenomenon of attracting moisture from the surrounding environment. Using the structure of zwitterion in the diagram below, show with the aid of a labelled diagram the hygroscopic nature of zwitterions. [2]



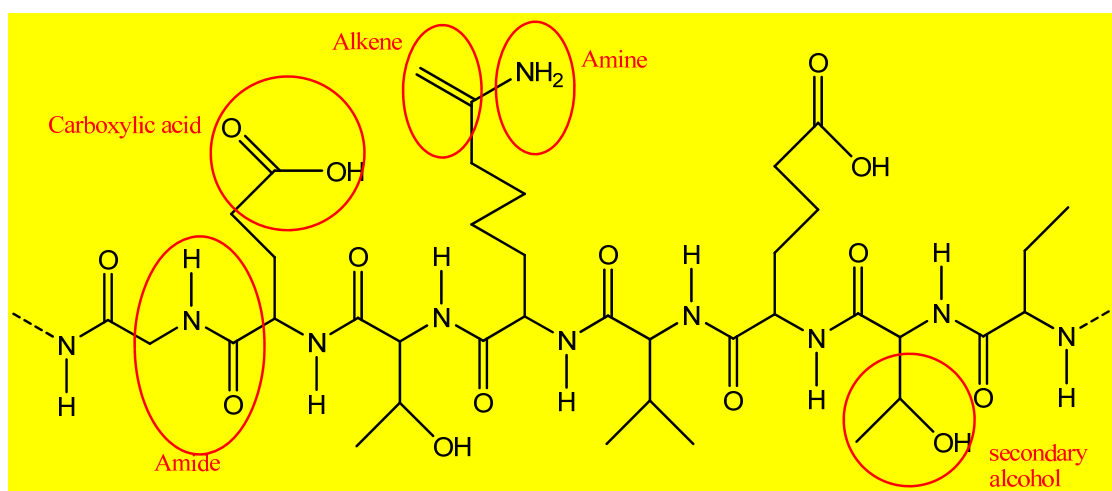
Ion-dipole interaction



The structure of a section of wool is shown below. This section consists of a number of amino acids.



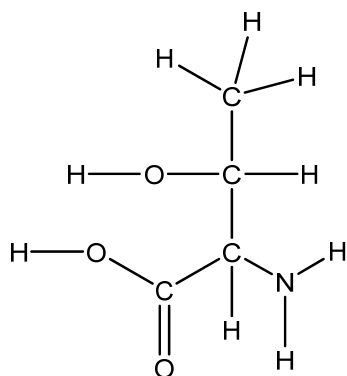
- (ii) On the structure above, **circle** and **name** 4 different functional groups present in wool. [2]



- (iii) How many amino acids are present in this portion of wool? [1]

8

- (iv) One of these amino acids are present more than once. Draw the displayed formula for this amino acid. [1]



- (d) Silver nanoparticles are used to colour wool fibres as well as to impart antimicrobial properties. Define the term nanoparticles and explain with reference to its structure, the advantages of using silver nanoparticles. [2]

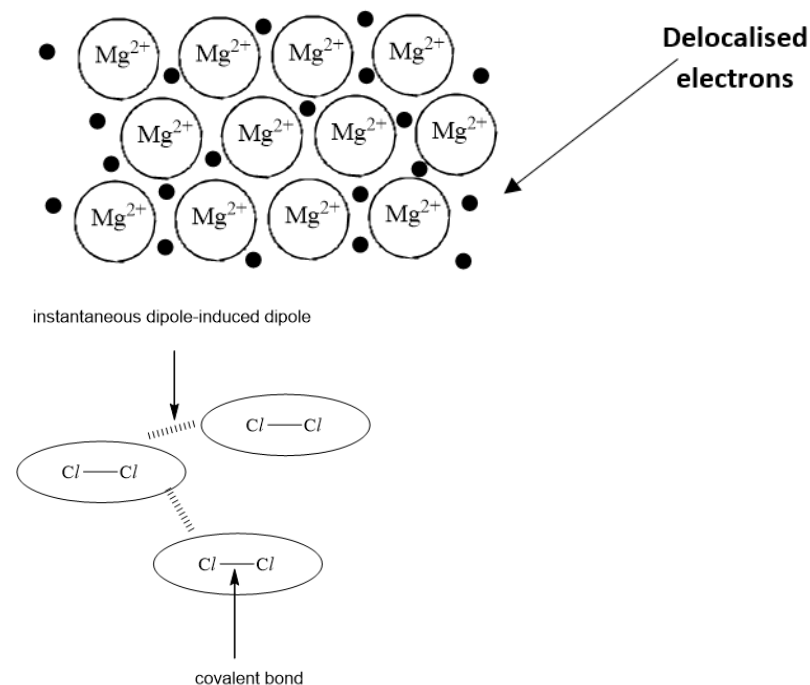
Nanoparticles are defined as particles with all dimensions between 1-100 nm on the nanoscale. The nanoparticle size helps to increase the surface area to volume ratio which increases the rate of the reaction.

[Total: 15]

Section B

Answer **one** question from this section, in the spaces provided.

- 1 (a) (i) State the type of bonding in magnesium and chlorine. Use a labelled diagram to illustrate each of their structure.



Words:

Mg: Metallic bonding

Cl₂: covalent bond (between Cl atoms/within the molecule)

instantaneous dipole-induced dipole between molecules

[4]

- (ii) Describe the structure of graphene.

Graphene has a single layer hexagonal lattice. Each carbon atom will overlap with 3 other carbon atoms to form 3 strong covalent bonds.

Each carbon atom also has an electron in the p orbital that can delocalize with neighbouring carbon atoms.

[2]

- (iii) State the similarities in the structure of magnesium and graphene that allow both of them to conduct electricity. Hence, state another physical property that both magnesium and graphene will have in common.

Both have delocalised electrons allowing both to also conduct heat

[2]

- 1 (b) Magnesium is an element in Period 3.

- (i) Define the term nucleon number.

[1]

Nucleon number is the total number of protons and neutrons in an atom.

In a particular experiment set-up, protons are deflected through an angle of $+15^\circ$.

- (ii) State one way in which the behaviour of electrons in an electric field differs from that of protons. [1]

Electrons are deflected to different plate/terminal from the protons (positive for electrons while negative for protons) OR electrons have a larger angle of deflection compared to protons.

- (iii) Calculate the angle of deflection of a Mg^+ particle. [1]

Angle of deflection = $(1/24.3) \times 15 = 0.617^\circ$

- (iv) Explain how the first ionisation energy of Mg would vary with the element before and after it. [3]

Mg has a higher nuclear charge than Na due to more protons

Increase in shielding effect is relatively constant as electron is added to the same quantum shell

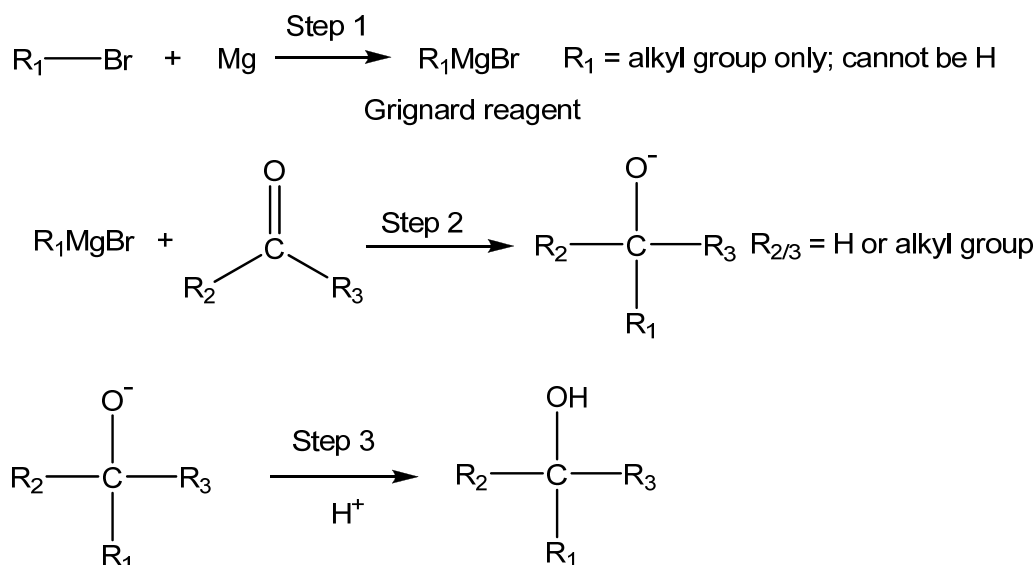
Effective nuclear charge larger for Mg and hence higher first IE

Valence electron for Al is in the 3p orbital which is further away from the nucleus

It also experience shielding from the 3s electrons

Outweighs the increase in nuclear charge hence first IE of Al is lower than Mg

- (c) Grignard reagents are formed by the reaction of magnesium metal with an organic compound. They can then be used to react with a carbonyl compound to obtain an alcohol as the final product. This is known as the Grignard reaction and can be represented by the following reaction scheme.



- (i) Name the functional group of the organic compound that reacted with magnesium to form the Grignard reagent in Step 1.

Halogenoalkane/Alkylhalide (alkylbromide)

[1]

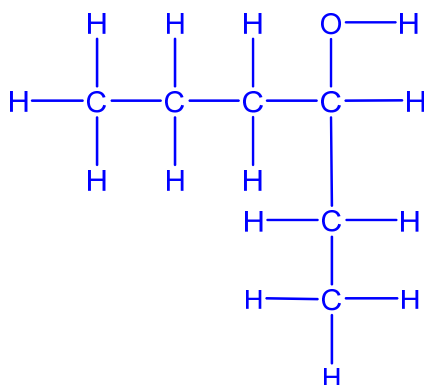
- 1 (c) (ii) Suggest the type of reaction that took place in Step 2.

Addition

[1]

- (iii) Draw the structural formula and give the IUPAC name of the alcohol that is formed when bromoethane was used to form the Grignard reagent and subsequently reacted with butanal.

Hexan-3-ol



[2]

- (iv) Butan-2-ol is an alcohol that can be obtained via the Grignard reaction. Draw the structure of a primary alcohol that is a constitutional isomer of butan-2-ol.



[1]

- (v) Suggest the functional group of the compound formed when carbon dioxide, CO_2 , is used in the Grignard reaction instead of a carbonyl compound.

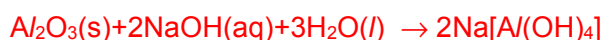
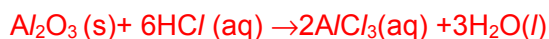
Carboxylic acid

[1]

[Total: 20]

- 2 Aluminium and sulfur are elements found in Period 3. When burnt in excess oxygen, their oxides, Al_2O_3 and SO_3 , will be produced.

- (a) Using relevant equations, justify the nature of both oxides.



[3]

- (b) (i) Draw the dot-and-cross diagram of SO_2 .

[1]



- (ii) Hence or otherwise, deduce whether SO_2 or SO_3 has a smaller bond angle. Explain your answer.

[1]

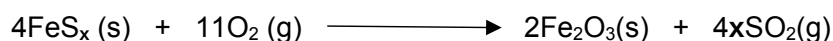
SO_2 has a smaller bond angle/ resulting in the bond pairs in SO_2 to be forced closer together.

This is due to lone-pair bond-pair repulsion in SO_2 is stronger than bond-pair bond pair repulsion in SO_3 , resulting in the bond pairs in SO_2 to be forced closer together.

Sulfuric acid (H_2SO_4) is a commonly used chemical reagent in many industrial processes and is manufactured through the Contact process. This takes place in 3 stages.

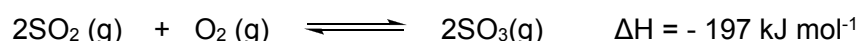
Stage 1:

In this stage, SO_2 is produced via burning pyrite, a compound containing only Fe and S, in excess oxygen. The equation at this stage is as follows:



Stage 2:

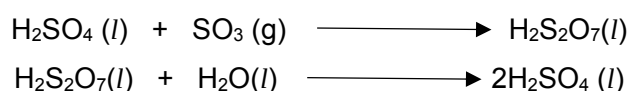
In this stage, the SO_2 produced is converted to SO_3 according to the following equation:



Stage 3:

In this stage, the SO_3 produced is converted to H_2SO_4 via 2 reactions. The first reaction involves dissolving the SO_3 produced in concentrated H_2SO_4 to obtain $\text{H}_2\text{S}_2\text{O}_7$.

The $\text{H}_2\text{S}_2\text{O}_7$ produced can then be safely added to water to produce H_2SO_4 in the second reaction. The equations at this stage are as follows:



To produce H_2SO_4 in a laboratory, a technician would first burn 10 g of pyrite in excess oxygen. The SO_2 produced would then be extracted and pumped into a 2 dm^3 vessel with more oxygen and allowed to react again at a moderately high temperature. When equilibrium has established, the SO_3 produced was quickly extracted and pumped into 50 cm^3 concentrated H_2SO_4 before the resultant solution was added to 50 cm^3 of deionised water to produce H_2SO_4 .

The H_2SO_4 produced was tested for its concentration where 25.0 cm^3 of the resultant solution required 23.50 cm^3 of 2.00 mol dm^{-3} NaOH for complete titration.

- (c) (i) Given that pyrite contained 53.5% of sulfur by mass, show that the value of x in FeS_x is 2. [1]

	Fe	S
%	46.5	53.5
/Ar	$46.5/55.8 = 0.833$	$53.5/32.1 = 1.667$
Smallest ratio	$0.833/0.833 = 1$	$1.667/0.833 = 2$

Since the formula is FeS_x

$x = 2$ (shown)

Using the information given above,

- (ii) Calculate the number of moles of SO₂ produced from the combustion of pyrite.

[1]

$$\text{No. of moles of pyrite} = 10/120 = 0.08333$$

$$\text{No. of moles of SO}_2 \text{ produced} = 0.08333 \times 2 = 0.167$$

- (iii) Calculate the total number of moles of H₂SO₄ obtained by the technician and hence the number of moles of SO₃ produced.

[2]

$$\text{No. of moles of NaOH} = (23.50/1000) \times 2 = 0.047$$

$$\text{No. of moles of H}_2\text{SO}_4 \text{ (25 cm}^3\text{)} = 0.047 / 2 = 0.0235$$

$$\text{No. of moles of H}_2\text{SO}_4 \text{ (100 cm}^3\text{)} = 0.0235 \times 4 = 0.094$$

$$\text{No. of moles of SO}_3 \text{ produced} = 0.094 / 2 = 0.0470$$

- (iv) Write an expression for the equilibrium constant, K_c, for the Contact process in stage 2.

[1]

$$K_c = [\text{SO}_3]^2 / ([\text{SO}_2]^2[\text{O}_2])$$

- (v) Using your answers obtained in 2(c)(ii) – (iv), complete Table 2.1.

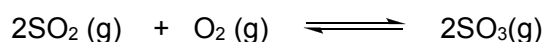


Table 2.1

	SO ₂	O ₂	SO ₃
initial amount/mol	0.167	1	0
change/mol	- 0.0470	-0.0235	+0.0470
equilibrium amount/mol	0.12	0.9765	0.0470

Hence or otherwise, calculate the value of K_c.

[2]

$$K_c = [\text{SO}_3]^2 / ([\text{SO}_2]^2[\text{O}_2])$$

$$= (0.0470/2)^2 / ((0.12/2)^2(0.9765/2))$$

$$= 0.3141$$

$$= 0.314 \text{ (mol}^{-1} \text{ dm}^3\text{)}$$

- (vi) To ensure a reasonable yield of SO_3 , the technician conducted the reaction between SO_2 and O_2 at moderately high temperatures. Explain why this is so.

[2]

Low temperature favours the production of SO_3 as position of equilibrium will shift to the right given the forward reaction is exothermic.

However, at low temperature, rate would be very slow

Temperature is kept moderately high so that the rate of the reaction is not too slow and yet favour a reasonable yield of SO_3 .

- (d) **X** has a molecular formula of $\text{C}_4\text{H}_{10}\text{O}_2$. 1 mole of **X** reacts with 1 mole of sodium to give 1 mole of hydrogen gas.

When acidified potassium manganate (VII) is heated with **X**, the solution turns colourless to give only **Y**.

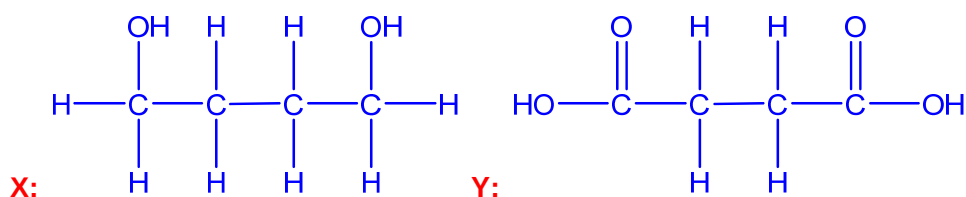
When 1 mole **Y** is added to methanol in the presence of concentrated sulfuric acid, **Z** is obtained together with 2 moles of water.

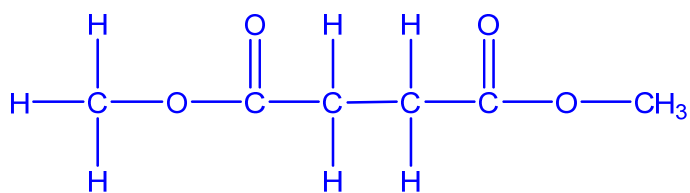
Deduce the structures of **X**, **Y** and **Z**, explaining the chemistry of the reactions described.

[6]

Observation	Deduction
1mole of X react with 1 mole of sodium to give 1 mole of hydrogen gas	X undergoes redox reaction with Na metal. Since 1 mole of X reacts with 1 mole of Na to give 1 mole of H_2 gas, it has 2 -OH/alcohol groups.
X reacts with acidified potassium manganate (VII) when heated to give only Y	X is oxidised to give Y X contained either primary or secondary alcohol Y contain carboxylic acids or ketone
1 mole of Y reacts with CH_3OH to give Z and 2 moles of water	Y undergo condensation [1/2] with CH_3OH to give Z . Since 2 moles of water are produced, 2 esters bonds in Z (diesters) Y contains 2 carboxylic acids

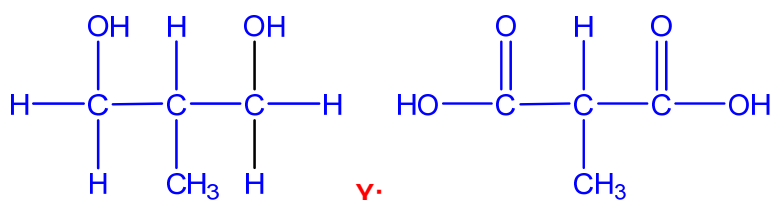
[5.5] max [3]





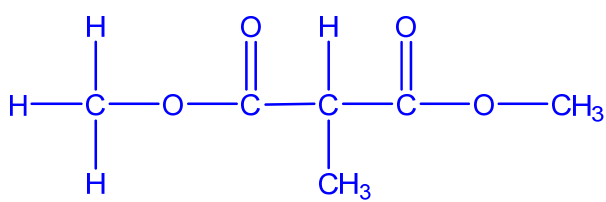
Z:

(alternative answer)



X:

Y:



Z:

END OF PAPER